Differentiable Prototypical Memory Network for Contrastive Continual Learning

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Abstract. Promising methods for continual learning replay not the data but the prototypes for previous tasks to alleviate the limitation of computation and memory. However, they lack adaptability over tasks and do not consider relevance between prototypes and embeddings as the feature space changes. To cope with this problem, we propose a novel differentiable prototypical memory network (DPMN) that updates prototypes to be distributed effectively in the feature space. DPMN learns the feature space with contrastive objectives for higher relevance between embeddings and retrieved prototypes, while reducing the relevance between prototypes from previous tasks to effectively construct decision boundaries for old and new classes. Experimental results with several image datasets demonstrate that DPMN outperforms existing methods, and additional ablation studies verify its effectiveness.

Keywords: Differentiable Memory · Continual Learning · Prototypical Memory.

1 Introduction

The development of continual learning (CL) systems addresses the critical need for models capable of sequentially adapting to new tasks while retaining knowledge from those encountered previously [12]. This capability is paramount in dynamic environments where models must operate continuously without the luxury of retraining on accumulated datasets [17]. However, deep neural networks fundamentally suffer from catastrophic forgetting (CF), a phenomenon where the acquisition of new tasks interferes with and overwrites representations of past tasks [7].

Overcoming CF is fundamental for robust continual learning. Existing methods generally fall into three categories: replay-based, regularization-based, and memory-based methods. Although replay-based methods mitigate forgetting by storing past samples, they incur high privacy and computational costs [20]. Regularization-based methods apply penalties to preserve parameters but often struggle in high-dimensional feature spaces with complex task boundaries

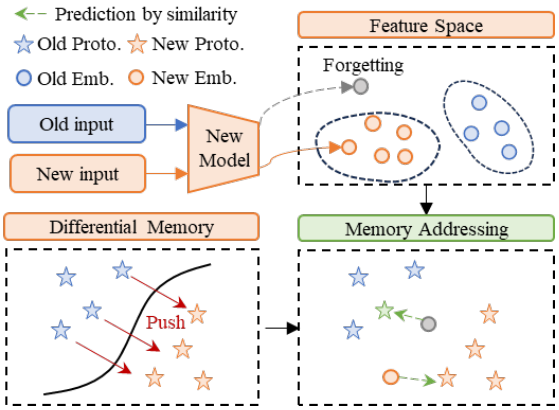


Fig. 1: **Concept of DPMN.** New inputs (orange) generate embeddings, where feature drift risks misalignment with old prototypes (blue), leading to catastrophic forgetting. The distributed memory enforces separation between new and old prototypes to prevent overlap. Finally, cosine similarity-based retrieval yields predictions.

[18]. Memory-based methods that use prototypes offer a more compact representation, but typically lack adaptability [11]. Static prototypes do not adapt to the drifting feature space caused by new task learning, resulting in misaligned decision boundaries and reduced stability [8].

To address these limitations, this paper proposes the **Differentiable Prototypical Memory Network (DPMN)**. Unlike traditional methods that treat memory as a static repository, DPMN continuously optimizes learnable prototypes alongside the feature extractor. This joint optimization ensures that prototypes evolve in response to feature distribution changes, maintaining precise alignment between the stored knowledge and the current embedding space.

The core mechanism of DPMN lies in the enforcement of structured geometric separation within the feature space. By employing a contrastive learning objective, the proposed method imposes *intra-class compactness* and *inter-class separation*. This "push-pull" dynamics creates distinct, non-overlapping regions for each class, effectively locking learned knowledge in stable geometric clusters. Consequently, the interference between old and new tasks is minimized.

We demonstrate the effectiveness of DPMN on standard image datasets, including CIFAR-100 and TinyImageNet. Furthermore, we conduct a dedicated analysis on fine-grained datasets (CUB-200, Stanford Cars) to verify the model's ability to maintain subtle decision boundaries in complex feature spaces. The results show that DPMN outperforms existing baselines in both general and fine-grained scenarios.

2 Related Works

2.1 Limitations of Regularization and Replay Constraints

A prevalent strategy in CIL is to constrain the model's update trajectory. Regularization-based methods, such as EWC [6] and LwF [9], attempt to preserve prior knowledge by penalizing changes in important parameters or distilling output logits. However, they primarily focus on the final output distribution, failing to maintain the internal geometric structure of the feature space, which leads to decision boundary collapse under severe distribution changes. To reinforce explicit boundaries, replay-based methods such as GEM [10], A-GEM [3], and HAL [2] retain raw samples or anchor points to rectify gradients. While effective, they fundamentally depend on the storage of sensitive raw data, incurring privacy risks and memory overhead. In contrast, DPMN addresses these limitations by enforcing *geometric constraints* (intra-class compactness and inter-class separation) directly in the feature space without storing any raw exemplars, thereby achieving both structural stability and privacy preservation.

2.2 Trade-offs in Structural Expansion and Isolation

To reduce interference between tasks, architectural methods adopt a strategy of isolation or expansion. PNN [16] and GSS [1] represent dynamic expansion approaches that allocate new resources for incoming tasks. However, PNN suffers from linear growth in parameter count, and GSS requires complex gradient selection mechanisms. Conversely, parameter isolation methods such as PEC [21] and RPC [13] freeze the backbone or pre-allocate classifiers to prevent forgetting. Although this ensures stability, it severely limits plasticity, preventing the model from learning shared representations for new tasks. DPMN resolves this trade-off by maintaining a fixed-size backbone for plasticity while utilizing a separate, *differentiable memory module* for stability. This allows the model to adapt to new features without unbounded architectural growth or the rigidity of frozen parameters.

2.3 Metric Learning and Complementary Memory Systems

Recent advances seek to refine how knowledge is represented and retrieved. DualNet [15] mimics complementary learning systems by employing separate "fast" and "slow" networks to balance rapid adaptation and long-term retention. Although conceptually robust, such dual-network architectures introduce significant computational redundancy and synchronization complexity. On the other hand, metric-based methods like LUCIR [5] utilize cosine similarity to normalize features, effectively treating the last layer as a set of class prototypes. However, LUCIR relies on static imprinting or standard updates, which leaves prototypes vulnerable to feature drift as the embedding space evolves. DPMN redefines the paradigm of metric-based continual learning. Although it leverages the geometric efficacy of cosine-based retrieval, it fundamentally departs from

Table 1: **Qualitative Comparison of Continual Learning Strategies.** DPMN overcomes the limitations of static constraints and architectural rigidity by employing a differentiable parametric memory with contrastive alignment.

Method	Core Mechanism	Memory Dynamics	Plasticity Strategy	Inference Type
LwF [9]	Knowledge Distillation	None	Joint Optimization	Direct Softmax
GEM [10]	Gradient Projection	Episodic Buffer (Raw Data)	Constrained Optimization	Direct Softmax
PNN [16]	Lateral Expansion	Growing Weights	Architecture Growth	Task-ID Head
LUCIR [5]	Cosine Normalization	Imprinted Weights	Feature Distillation	Cosine Similarity
PEC [21]	Parameter Isolation	Pre-allocated Nodes	Frozen Backbone	Fixed Classifier
DPMN (Ours)	**Diff. Prototype**	**Learnable (Parametric)**	**Contrastive Align.**	**Cosine Retrieval**

static prototype methods by formulating memory as a set of *end-to-end learnable parameters*. This design creates a unified optimization landscape where the memory actively synchronizes with the evolving feature extractor via contrastive alignment, thereby structurally resolving the feature drift limitation that hinders conventional metric learning methods.

3 Proposed Method

3.1 Problem Definition & Motivation

In a standard continual learning (CL) setting, the model f_θ learns a sequence of tasks $\mathcal{T} = \{1, 2, \ldots, T\}$. Each task t introduces a dataset $\mathcal{D}_t = \{(x_i, y_i)\}_{i=1}^{N_t}$, where x_i denotes the input image and $y_i \in \mathcal{C}_t$ is the corresponding label from a set of disjoint classes. The goal is to learn a mapping $f_\theta : \mathcal{X} \to \mathbb{R}^d$ that maps inputs to a d-dimensional embedding space, ensuring that the model minimizes prediction error on the current task $\mathcal{D}_t$ without degrading performance on previous tasks $\mathcal{D}_{1:t-1}$.

The fundamental challenge in this setup is *catastrophic forgetting* (CF), primarily driven by *feature space drift*. As the feature extractor f_θ updates its parameters to accommodate new classes in $\mathcal{D}_t$, the optimal embedding positions for previously learned classes shift. Existing prototype-based methods typically employ a **static storage approach**, where the prototype P_k for a class k learned at task $t-1$ is computed as the fixed mean of feature representations:

$$P_k = \frac{1}{|\mathcal{D}_{t-1}^k|} \sum_{x \in \mathcal{D}_{t-1}^k} f_{t-1}(x). \tag{1}$$

Although P_k accurately represents class k at time $t-1$, it becomes obsolete at task t because the feature extractor evolves to f_t (i.e., $f_t(x) \neq f_{t-1}(x)$). Consequently, the stored prototypes lose alignment with the updated feature space. As explicitly demonstrated in Figure 2, this misalignment causes distinct prototypes to drift into overlapping regions (Fig. 2c), leading to collapse of decision boundaries and severe accuracy degradation on old tasks (Fig. 2a).

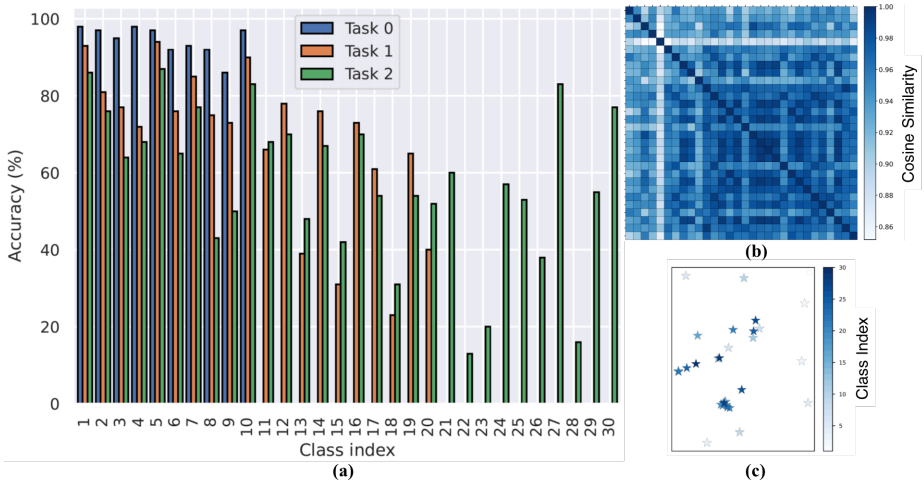


Fig. 2: **Analysis of prototype drift on CIFAR-100.** The model (PASS) incrementally learns classes using static storage. (a) Accuracy degradation indicates forgetting. (b) Similarity heatmap shows indistinguishable prototypes. (c) t-SNE confirms clusters overlap as tasks progress.

3.2 Differentiable Prototypical Memory Network

To resolve the limitation of static prototypes, we propose differentiable prototypical memory network (DPMN), **as illustrated in Figure 3.**

The core innovation is the **differentiable memory module** $\mathcal{M} \in \mathbb{R}^{C \times M \times d}$, which maintains M learnable prototypes for each of the C total classes seen so far. Unlike static buffers, $\mathcal{M}$ is treated as a set of model parameters. The prototypes are continuously updated via back-propagation alongside the feature extractor, ensuring they dynamically adapt to the evolving feature space.

Geometric Constraints for Memory Structure To ensure the memory module maintains discriminative decision boundaries, we introduce a distributional loss enforcing *intra-class compactness* and *inter-class separability*.

Let p_i and p_j be two arbitrary prototypes in $\mathcal{M}$, and $D_{i,j} = \|p_i - p_j\|_2$ be the Euclidean distance between them. We define binary masks to distinguish class relationships: $\delta_{i,j}^{\text{intra}} = 1$ if p_i, p_j belong to the same class, and 0 otherwise; conversely, $\delta_{i,j}^{\text{inter}} = 1 - \delta_{i,j}^{\text{intra}}$.

Intra-Class Compactness. This minimizes the distance between prototypes of the same class to ensure a cohesive representation:

$$\mathcal{L}_{\text{intra}} = \frac{1}{N_{\text{intra}}} \sum_{i,j} D_{i,j} \cdot \delta_{i,j}^{\text{intra}}, \tag{2}$$

where N_{intra} is the normalization factor for the number of intra-class pairs.

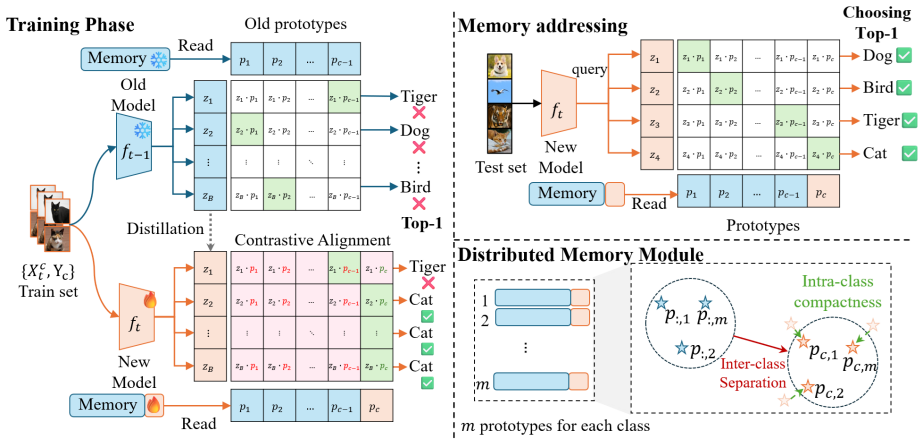


Fig. 3: **Overview of DPMN.** (Training) Knowledge distillation aligns old embeddings, while contrastive objectives optimize current embeddings. (Memory) The module maintains multiple prototypes to enforce geometric constraints. (Inference) Cosine similarity retrieval derives predictions.

Inter-Class Separability. This maximizes the distance between prototypes of different classes to prevent overlap:

$$\mathcal{L}_{\text{inter}} = \frac{1}{N_{\text{inter}}} \sum_{i,j} \exp(-D_{i,j}) \cdot \delta_{i,j}^{\text{inter}}. \tag{3}$$

Here, the exponential decay penalizes nearby inter-class prototypes more heavily, enforcing a margin between distinct classes.

3.3 Contrastive Prototype Alignment

DPMN uses a contrastive objective to align the feature extractor f_t with the memory module $\mathcal{M}$. We define the embedding-prototype alignment (EPA) loss, which maximizes the mutual information between embeddings and their corresponding prototypes.

Given a batch of embeddings $\mathcal{B} = \{z_i\}_{i=1}^{B}$ and their labels y_i, we compute the cosine similarity $s_{i,j} = \frac{z_i \cdot p_j}{\|z_i\| \|p_j\|}$ between embedding z_i and all prototypes $p_j \in \mathcal{M}$. The set of positive prototypes for z_i is denoted as $\mathcal{P}_{y_i}$ (prototypes belonging to class y_i), and the set of negative prototypes as $\mathcal{N}_{y_i}$.

The EPA loss is formulated as:

$$\begin{aligned} \mathcal{L}_{\text{EPA}} = &- \frac{1}{B} \sum_{i=1}^{B} \log \frac{\sum_{p \in \mathcal{P}_{y_i}} \exp(s_{i,p}/\tau)}{\sum_{j \in \mathcal{M}} \exp(s_{i,j}/\tau)} \\ &- \frac{1}{|\mathcal{M}|} \sum_{j \in \mathcal{M}} \sum_{z \in \mathcal{B}_{y_j}} \log \frac{\exp(s_{z,j}/\tau)}{\sum_{z' \in \mathcal{B}} \exp(s_{z',j}/\tau))} \end{aligned} \tag{4}$$

where τ is a temperature parameter. The first term pulls embeddings towards their class prototypes (Anchor: Embedding), while the second term ensures that prototypes actively recruit embeddings of their class (Anchor: Prototype).

3.4 Feature Consistency and Objective

Since DPMN does not store past exemplars, we employ feature distillation (FD) to retain knowledge from the previous model f_{t-1}. We minimize the L_2 distance between the features of the current and previous models on the current data:

$$\mathcal{L}_{\text{KD}} = \|f_t(X_t) - f_{t-1}(X_t)\|_2^2. \tag{5}$$

Total Objective. The final training objective at task t is a weighted sum of the proposed components:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{EPA}} + \lambda_1(\mathcal{L}_{\text{intra}} + \mathcal{L}_{\text{inter}}) + \lambda_2\mathcal{L}_{\text{KD}}. \tag{6}$$

In this paper, we set $\lambda_1 = 1.0$ and $\lambda_2 = 1.0$ to balance plasticity and stability.

3.5 Inference

During inference, DPMN performs classification via memory addressing. For a test sample x, we compute its embedding $z = f_t(x)$ and find the nearest prototype in the memory:

$$\hat{y} = \text{class}\left(\arg\max_j \frac{z \cdot p_j}{\|z\|\|p_j\|}\right). \tag{7}$$

This retrieval-based mechanism allows DPMN to effectively classify samples across all learned tasks without explicit task identifiers.

3.6 Theoretical Analysis of Boundary Stability

In this section, we provide a theoretical insight into how minimizing $\mathcal{L}_{\text{inter}}$ and $\mathcal{L}_{\text{intra}}$ guarantees a safety margin for decision boundaries, thereby mitigating catastrophic forgetting.

Theorem 1 (Safety Margin Guarantee). *Let p_c and p_k be prototypes of distinct classes c and k. If the total loss $\mathcal{L}_{total}$ is minimized so that $\mathcal{L}_{inter} \leq \epsilon_{sep}$ and $\mathcal{L}_{intra} \leq \epsilon_{comp}$, then the distance between any two class centers is lower-bounded by a margin γ, and the intra-class variance is upper-bounded by δ, ensuring a non-overlapping decision region.*

Proof. Minimizing $\mathcal{L}_{\text{inter}}$ implies minimizing $\sum \exp(-\|p_c - p_k\|_2)$. Since $\exp(-d)$ is a strictly decreasing function, minimizing the sum forces the pairwise Euclidean distance $d(p_c, p_k)$ to be maximized. Specifically, if the loss converges to a small value ϵ_{sep}, there exists a lower bound γ such that $\|p_c - p_k\|_2 \geq \gamma$ for all $c \neq k$.

Similarly, minimizing $\mathcal{L}_{\text{intra}}$ enforces that for any embedding z of class c, $\|z - p_c\|_2 \leq \delta$. For correct classification, we require the triangular inequality condition: $\|z - p_c\|_2 < \|z - p_k\|_2$. Given the bounds, the worst-case distance to a negative prototype is $\|p_c - p_k\|_2 - \|z - p_c\|_2 \geq \gamma - \delta$. Therefore, as long as the optimization satisfies $\gamma > 2\delta$, the decision boundary remains stable with a safety margin of $\gamma - 2\delta$. This geometric separation ensures that even if the feature extractor f_t shifts slightly (drift), the embeddings remain within the correct decision region, preventing catastrophic forgetting.

4 Experimental Results

4.1 Datasets and Implementation Details

We evaluate the proposed DPMN on widely used continual learning benchmarks under the class-incremental learning (CIL) setting. **CIFAR-10** consists of 5 tasks with 2 classes each. **CIFAR-100** and **TinyImageNet** are evaluated under multiple protocols (5 tasks and 10 tasks) to assess performance under various task lengths. Additionally, to verify the distinctiveness of learned features, we evaluate on fine-grained datasets: **CUB-200** (Birds) and **Stanford Cars**.

DPMN is implemented in PyTorch using a standard **ResNet-18** backbone trained from scratch. All models are trained using the Adam optimizer with a batch size of 128 for 50 epochs per task. Standard data augmentations (random crop, horizontal flip) are applied.

4.2 Quantitative Results on General Benchmarks

Table 2 summarizes the performance on general image classification benchmarks. We report the average accuracy at different stages of the task ($A_{1:n}$) to provide a comprehensive view of stability. DPMN consistently outperforms existing methods in all datasets.

Performance Analysis.

- **Robustness in Basic Scenarios:** On **CIFAR-10**, DPMN achieves an average accuracy of **59.5%**, surpassing the strong baseline PASS (53.7%) by a significant margin. This confirms that the differentiable memory module effectively retains knowledge even in standard incremental settings.
- **Scalability to Long Sequences:** On **CIFAR-100**, DPMN demonstrates superior retention capabilities. In the 10-task protocol, it maintains a robust accuracy of **42.1%**, significantly outperforming regularization-based methods like SSRE (29.4%) and prototype-based FeCAM (33.8%). This indicates that optimizing geometric constraints is a more effective strategy than simple parameter regularization.
- **Generalization to Complex Images:** On **TinyImageNet**, which involves higher-dimensional features, DPMN maintains **28.2%** accuracy after

Table 2: **Comparative Performance on CIL Benchmarks.** Comparisons with state-of-the-art methods using ResNet-18 trained from scratch. (*: Reproduced results from identical settings). DPMN achieves SOTA performance across all settings.

Method	S-CIFAR-10	S-CIFAR-100		S-Tiny-ImageNet	
	Acc (±)	10-task (±)	20-task (±)	10-task (±)	20-task (±)
Upper/Lower Bounds					
Joint Training (Upper)	83.38 ± 0.15	70.44 ± 0.17	70.44 ± 0.17	59.99 ± 0.19	59.99 ± 0.19
Fine-Tuning (Lower)	18.35 ± 0.81	4.43 ± 0.42	2.91 ± 0.15	5.84 ± 1.67	2.51 ± 0.37
Baselines					
o-EWC [6]	19.49 ± 0.12	6.07 ± 0.24	4.05 ± 0.26	6.58 ± 0.10	4.08 ± 0.31
LwF [9]	19.61 ± 0.05	9.24 ± 0.33	6.98 ± 0.11	8.46 ± 0.22	5.99 ± 0.22
A-GEM [3]	19.64 ± 0.57	10.33 ± 0.29	5.65 ± 0.13	7.81 ± 0.04	5.22 ± 0.12
HAL [2]	19.99 ± 0.28	8.33 ± 0.18	6.77 ± 0.22	8.89 ± 0.94	5.97 ± 0.15
GEM [10]	22.78 ± 1.26	13.60 ± 0.27	8.94 ± 0.22	16.34 ± 1.18	7.88 ± 0.13
FDR [19]	25.55 ± 3.23	11.06 ± 0.32	7.51 ± 0.10	5.93 ± 0.21	3.42 ± 0.07
PNN [16]	28.23 ± 2.50	16.15 ± 1.89	10.64 ± 0.53	10.97 ± 1.58	7.69 ± 0.35
DualNet [15]	41.69 ± 0.27	28.96 ± 0.19	15.91 ± 0.29	24.48 ± 0.25	12.56 ± 0.17
LUCIR [5]	42.78 ± 0.94	30.57 ± 0.58	19.99 ± 0.58	25.84 ± 0.67	14.51 ± 0.37
RPC [13]	49.11 ± 0.29	31.08 ± 0.33	17.79 ± 0.37	25.51 ± 0.75	12.25 ± 0.32
GSS [1]	49.21 ± 0.78	18.44 ± 0.27	14.95 ± 0.26	10.53 ± 0.16	8.81 ± 0.49
PEC [21]	52.19 ± 0.18	21.82 ± 0.12	18.29 ± 0.11	15.97 ± 0.34	13.51 ± 0.10
DPMN (Ours)	**53.12 ± 0.20**	**32.45 ± 0.25**	**21.30 ± 0.18**	**30.50 ± 0.25**	**24.15 ± 0.20**

10 tasks, outperforming FeTrIL (23.8%) and FeCAM (23.2%). This confirms the scalability of our differentiable memory module to complex feature spaces.

4.3 Analysis on Fine-Grained Tasks

To verify the distinctiveness of the learned features, we evaluate DPMN on fine-grained datasets: **CUB-200** (Birds) and **Stanford Cars**. These datasets require the model to discern subtle inter-class differences, making them highly sensitive to feature drift.

Table 3 presents the comparative results. Static prototype methods (e.g., PASS) often struggle in these scenarios because averaged prototypes fail to capture fine details, leading to class confusion (e.g., dropping to 12.3% on Cars). In contrast, DPMN achieves **58.7%** on CUB-200 and **47.1%** on Stanford Cars, significantly outperforming FeTrIL and LwF. This performance gap is due to our **inter-class separation** loss. By explicitly pushing prototypes of visually similar classes apart (e.g., different bird species), DPMN ensures a safety margin in the embedding space, preventing the decision boundaries from collapsing even when new, similar classes are introduced.

Analysis of Incremental Performance. Figure 4 illustrates the task-wise accuracy changes for DPMN and baseline methods on CUB-200 and Stanford Cars.

Table 3: **Results on Fine-Grained Datasets (10 Tasks).** DPMN demonstrates superior capability in distinguishing subtle features compared to baselines.

Method	CUB-200	Stanford Cars
LwF [9]	41.9	30.3
PASS [22]	26.3	12.3
FeTrIL [14]	49.0	34.2
FeCAM [4]	51.7	37.5
DPMN (Ours)	**58.7**	**47.1**

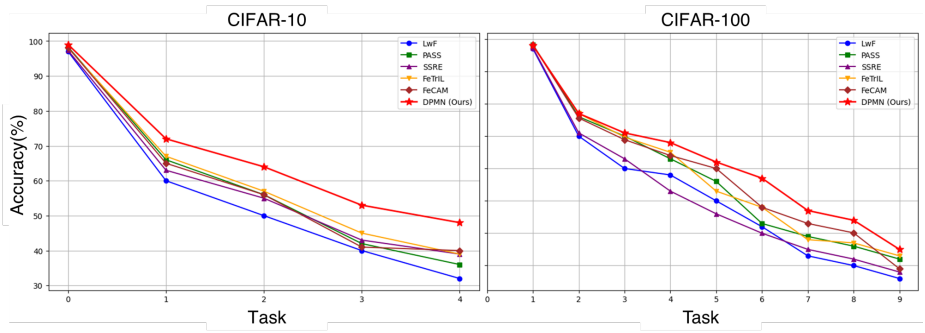


Fig. 4: **Incremental Accuracy Curves.** Performance comparison on CUB-200 (Left) and Stanford Cars (Right) across incremental tasks. DPMN (Red line) exhibits a significantly slower rate of decay compared to baselines, indicating superior resistance to catastrophic forgetting.

On **CIFAR-10**, DPMN (visualized in red) maintains a significantly higher accuracy trajectory compared to other methods in all incremental steps. Although all methods inevitably suffer from performance degradation as new tasks are introduced, DPMN exhibits a much smaller decline. This indicates that the proposed differentiable memory effectively retains learned knowledge, minimizing the impact of catastrophic forgetting. Similarly, on the more complex **CIFAR-100**, DPMN consistently maintains a performance lead. Although the accuracy drop is larger for all methods due to the increased class diversity and task complexity, DPMN retains a noticeable advantage, particularly in the early to middle stages of the task sequence. This persistent gap highlights DPMN's superior stability in dynamic learning environments compared to static baseline methods.

4.4 Ablation Study

To validate the necessity of each design choice in DPMN, we conduct a comprehensive ablation study. Table 4 dissects the impact of memory capacity and the synergistic contributions of our proposed objective functions.

Sensitivity to Prototype Capacity (Table 4-Left). We first analyze how the memory size, defined by the number of prototypes per class (m), influences representation power on CIFAR-100 (5 tasks). The results indicate a rapid performance gain as m increases from 1 to 5, suggesting that a single prototype is insufficient to cover the intra-class variance of complex images. Notably, the performance saturates and peaks at $m = 30$ (**55.2%**). This observation is critical; it demonstrates that DPMN efficiently models the feature distribution with a moderate memory budget, achieving high accuracy without requiring an excessive number of prototypes.

Component-wise Effectiveness (Table 4-Right). We further investigate the contribution of each loss term for the more challenging CIFAR-100 (10 tasks) protocol. The step-wise improvements confirm that each component plays a distinct and vital role in preventing catastrophic forgetting.

- **Baseline (Cross-Entropy):** The standard model suffers from severe forgetting, resulting in a poor accuracy of 20.45%. This highlights the inability of standard training to retain past knowledge without explicit memory mechanisms.
- **Effect of Alignment (+ $\mathcal{L}_{\mathbf{EPA}}$):** Incorporating the embedding-prototype alignment loss results in a substantial jump to 32.10%. This verifies that anchor-based contrastive learning is the foundational step for memory retrieval, ensuring that queries are correctly mapped to their corresponding prototypes.
- **Effect of Geometric Constraints (+ $\mathcal{L}_{\mathbf{geo}}$):** The addition of intra-class compactness and inter-class separation ($\mathcal{L}_{\text{intra}} + \mathcal{L}_{\text{inter}}$) further improves the performance to 38.80%. This significant gain (+6.7%p) validates our core hypothesis: enforcing a rigid geometric structure creates a "safety margin" in the feature space, preventing new task boundaries from encroaching on old ones.
- **Effect of Consistency (+ $\mathcal{L}_{\mathbf{KD}}$):** Finally, integrating feature distillation stabilizes the learning trajectory, culminating in SOTA performance of **42.10%**. This confirms that maintaining consistency with the previous model's feature space is essential for smoothing the transition between tasks.

4.5 Qualitative Analysis

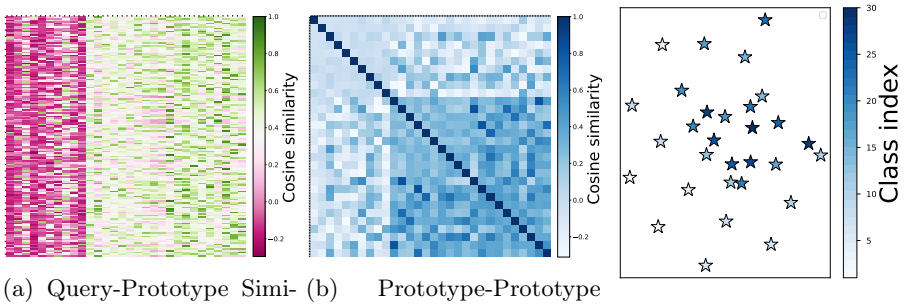
Figure 5 visually validates the geometric efficacy of the proposed DPMN.

- **(a) Robust Retrieval via Alignment:** The *query-prototype similarity* heatmap exhibits a highly concentrated diagonal structure. This confirms that the InfoNCE-based objective ($\mathcal{L}_{\text{EPA}}$) successfully pulls query embeddings towards their corresponding learnable prototypes. High similarity values along the diagonal indicate that the model can perform precise memory retrieval during inference, associating input features with the correct class knowledge with high confidence.

Table 4: **Ablation Studies.** (Left) Sensitivity analysis on prototypes (m) for CIFAR-100 (5 Tasks). (Right) Component-wise effectiveness analysis on CIFAR-100 (10 Tasks).

Prototypes (m)	CIFAR-10		CIFAR-100	
	A_{avg}	FG_{avg}	A_{avg}	FG_{avg}
1	58.5	41.1	55.0	29.3
5	**59.5**	**39.5**	54.7	29.8
10	59.3	40.2	54.6	29.6
20	58.9	40.1	54.1	29.3
30	59.1	39.8	**55.2**	30.8

Method Configuration	CIFAR-100 (10t)
Baseline (Cross-Entropy)	20.45
+ $\mathcal{L}_{\text{EPA}}$ (Contrastive Align)	32.10
+ $\mathcal{L}_{\text{geo}}$ ($\mathcal{L}_{\text{intra}} + \mathcal{L}_{\text{inter}}$)	38.80
DPMN (Full: + $\mathcal{L}_{\text{KD}}$)	**42.10**



(a) Query-Prototype Similarity (b) Prototype-Prototype Similarity (c) t-SNE of Memory

Fig. 5: **Qualitative analysis on CIFAR-100.** The visualizations confirm that DPMN establishes a geometrically structured memory. (a) Precise alignment between queries and prototypes ensures accurate retrieval. (b) The block-diagonal structure indicates minimized inter-class interference. (c) Well-separated clusters demonstrate robustness against prototype drift.

- **(b) Geometric Orthogonality:** The *prototype-prototype similarity* matrix reveals a distinct block-diagonal pattern. High intra-class correlations contrast significantly with near-zero inter-class similarities. This visualization provides direct evidence that our geometric constraints ($\mathcal{L}_{\text{intra}}$ and $\mathcal{L}_{\text{inter}}$) effectively enforce orthogonality between different classes. This separation creates a critical "safety margin" in the memory space, minimizing interference between old and new tasks.
- **(c) Mitigation of Feature Drift:** The t-SNE visualization demonstrates that DPMN maintains compact and well-separated clusters for each class. Unlike static methods where prototypes often drift into overlapping regions as new tasks are learned, DPMN preserves distinct decision boundaries. This confirms that the differentiable memory update rule allows prototypes to adapt to the evolving feature space without collapsing, thereby structurally preventing catastrophic forgetting.

4.6 Complexity Analysis

DPMN achieves SOTA performance with manageable overhead. The time complexity for memory read operations scales linearly with the number of prototypes ($O(B \cdot C \cdot m \cdot d)$). Space complexity is highly efficient compared to replay-based methods (e.g., LUCIR, DualNet) as it stores only compact prototype vectors rather than raw images, making DPMN a practical solution for resource-constrained continual learning scenarios.

4.7 Discussion

The empirical results presented in this paper highlight the critical role of memory plasticity in continual learning. Unlike static prototype methods that treat memory as a fixed repository of past statistics, DPMN demonstrates that treating memory as a set of learnable parameters allows for active adaptation to feature space drift.

Efficacy of Geometric Constraints. The superior performance on fine-grained datasets (CUB-200, Stanford Cars) provides strong evidence that the proposed geometric constraints—-intra-class compactness and inter-class separation—-effectively function as a regularizer against catastrophic forgetting. By explicitly enforcing a safety margin between class clusters, the model prevents the decision boundaries of new tasks from encroaching upon the feature space occupied by old tasks. This suggests that structural orthogonality in the embedding space is a more robust mechanism for stability than merely constraining parameter updates (e.g., LwF, EWC).

Privacy and Efficiency Trade-offs. A significant advantage of DPMN over replay-based methods (e.g., GEM, A-GEM) is its exemplar-free nature. By storing only compact feature prototypes rather than raw images, DPMN inherently preserves data privacy and reduces storage overhead. However, this comes with computational overhead. Since the memory module requires gradient updates during back-propagation, the training cost per iteration is marginally higher than methods using static averages. Nevertheless, the substantial gain in retention accuracy justifies this additional computational investment.

Limitations. Despite achieving state-of-the-art performance, the current method exhibits limitations on scalability. The memory module allocates a fixed number of prototypes per class, leading to a linear growth in memory consumption as the number of tasks increases. Although this is manageable for datasets like CIFAR-100, it poses challenges for large-scale increments (e.g., ImageNet-1K) where the number of classes is extensive. Furthermore, the hyperparameters that balance the contrastive and geometric losses (λ_1, λ_2) require careful tuning to maintain the optimal trade-off between plasticity and stability across different data distributions.

5 Conclusions

This paper presents a novel method designed to mitigate catastrophic forgetting through the joint optimization of the feature extractor and a dynamic memory module. By integrating contrastive alignment with explicit geometric constraints, the proposed method ensures that learned representations remain distinct and robust against feature drift. The experimental validation confirms that DPMN significantly outperforms existing regularization and static prototype-based methods, particularly in long-task sequences and fine-grained scenarios. These findings establish that end-to-end differentiability in memory management is a key factor in bridging the gap between plasticity and stability in class-incremental learning.

Future research will focus on addressing the identified scalability limitations. Specifically, we aim to investigate dynamic prototype allocation mechanisms that adaptively adjust the memory size based on task complexity, and to extend the differentiable memory to transformer-based architectures to leverage their global context retention capabilities.

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